MySQL Assignment 2 – Querying Data

1. **Distinct Values:**

    ⦿ a query to retrieve distinct salaries from the Employees table.

```
SELECT DISTINCT salary FROM employees;
```

2. **Alias (AS):**

    ⦿ Provide aliases for the "age" and "salary" columns as "Employee_Age" and "Employee_Salary", respectively.

```
SELECT age as Employee_Age, salary as Employee_Salary FROM employees;
```

3. **Where Clause & Operators:**

    ⦿ Retrieve employees with a salary greater than ₹50000 and hired before 2016-01-01.

```
SELECT * from employees where salary > 50000 and hire_date < '2016-01-01';
```

    ⦿ Find the employee whose designation is missing and fill it with "Data Scientist".

```
SELECT * FROM employees WHERE designation IS NULL;

UPDATE employees SET designation = 'Data Scientist' WHERE designation IS NULL;
```

Sorting and Grouping Data:

1. **ORDER BY:**

    ⦿ Find employees sorted by department ID in ascending order and salary in descending order.

SELECT * FROM employees ORDER BY department_id ASC, salary DESC;

2. **LIMIT:**

    ⦿ Display the first 5 employees hired in the year 2018.

SELECT * FROM employees WHERE hire_date >= ’2018-01-01’ AND
hire_date < ‘2019-01-01’ORDER BY hire_date ASC LIMIT 5;

3. **Aggregate Functions:**

    ⦿ Calculate the sum of all salaries in the Finance department.

SELECT SUM(e.salary) AS total_finance_salary FROM employees e JOIN departments d ON e.department_id = d.department_id WHERE d.department_name = 'Finance';

    ⦿ Find the minimum age among all employees.

SELECT MIN(age) AS minimum_age FROM employees;

4. **GROUP BY:**

    ⦿ List the maximum salary for each location.

SELECT l.location, MAX(e.salary) AS maximum_salary FROM employees e JOIN location l ON e.location_id = l.location_id GROUP BY l.location;

    ⦿ Calculate the average salary for each designation containing the word 'Analyst'.

SELECT designation, AVG(salary) AS average_salary FROM employees WHERE designation LIKE '%Analyst%' GROUP BY designation;

5. **HAVING:**

    ⦿ Find departments with less than 3 employees.

SELECT d.department_id, d.department_name, COUNT(e.employee_id) AS employee_count FROM departments d LEFT JOIN employees e ON d.department_id = e.department_id GROUP BY d.department_id, d.department_name HAVING COUNT(e.employee_id) < 3;

    ⦿ Find locations with female employees whose average age is below 30.

SELECT l.location, AVG(e.age) AS average_age FROM employees e JOIN location l ON e.location_id = l.location_id WHERE e.gender ='F' GROUP BY l.location HAVING avg(e.age) < 30;

Joins:

1. **Inner Join:**

    ⦿ List employee names, their designations, and department names where employees are assigned to a department.

SELECT e.employee_name, e.designation, d.department_name FROM employees e INNER JOIN departments d ON e.department_id = d.department_id;

2. **Left Join:**

    ⦿ List all departments along with the total number of employees in each department, including departments with no employees.

SELECT d.department_id, d.department_name, COUNT(e.employee_id) AS total_employees FROM departments d LEFT JOIN employees e ON d.department_id = e.department_id GROUP BY d.department_id, d.department_name;

3. **Right Join:**

    ⦿ Display all locations along with the names of employees assigned to

each location. If no employees are assigned to a location, display NULL for employee name.

```
SELECT l.location_id, l.location, e.employee_name  FROM employees e RIGHT JOIN location l ON e.location_id = l.location_id;
```